

# Inventory Analysis with Power BI

CASE STUDY: INVENTORY ANALYSIS IN POWER BI



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# What is a case study?

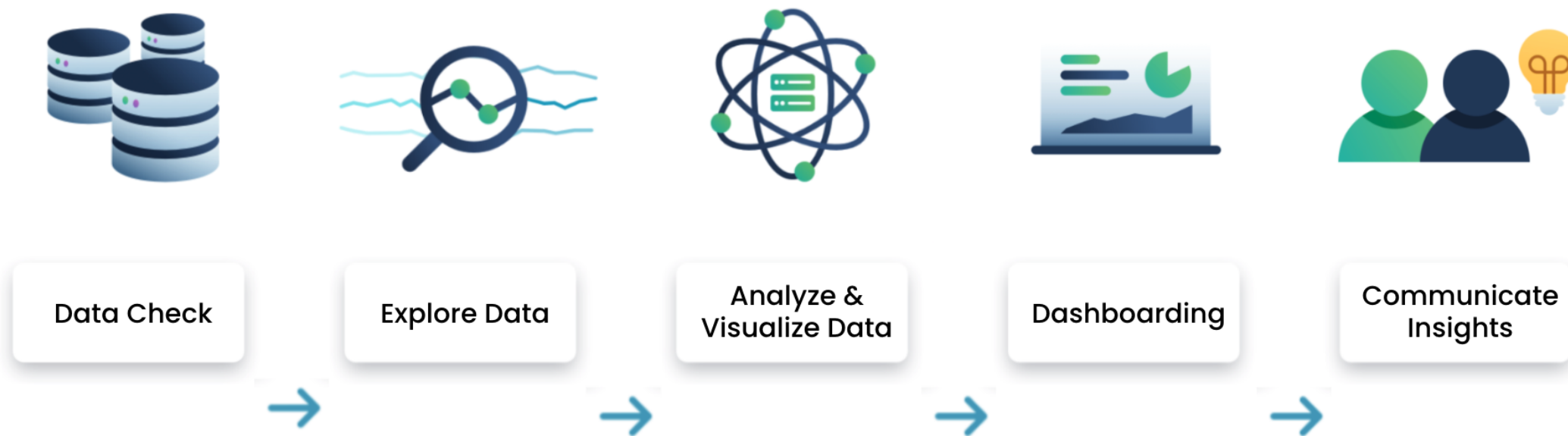
- Examination of a particular real-world problem
- Practice your skills
- No new Power BI concepts

## Prerequisite courses

- Introduction to Power BI
- Data Preparation in Power BI
- DAX Functions in Power BI



# Steps of the analysis



- 1) Verify the necessary data is present and clean
- 2) Explore the data
- 3) Analyze and visualize information
- 4) Create dashboards and report insights

# The problem

Help a management team make the best decisions.

- **WarmeHands Inc.**
- Objective: to communicate insights for the renewal of items



# What is an inventory analysis?

- Optimizing the management of inventory
- Some benefits of inventory analysis:
  - Reduction of storage expenses
  - Reduction of purchase costs
  - Reduction of unused goods
  - Helps to increase profit



# Starting with the inventory analysis

Some important calculations:

- Revenue
- Cost of Goods Sold (COGS)
- Profit

# Calculating revenue

Revenue is the money generated from normal business operations, calculated as the sales price times the number of units sold

$$\text{Revenue} = \text{quantity of units sold} \times \text{selling price}$$

<sup>1</sup> <https://www.investopedia.com/terms/r/revenue.asp>

# Cost of goods sold

Cost of producing or acquiring products that a company sells, including:

- Raw materials
- Labor
- Factory overhead or costs related to production facilities

## Example

*E.g. Wooden chair COGS = wood + glue + wood finishers + labor*

<sup>1</sup> <https://www.irs.gov/publications/p535>

# Exploring gross profit

Profit a company makes after deducting the costs associated with making and selling its products

Simplified formula

$$\textit{Gross Profit} = \textit{Revenue} - \textit{COGS}$$

<sup>1</sup> <https://www.investopedia.com/terms/g/grossprofit.asp>

# Dataset

Six tables with information on item description, initial stock, orders, dates, countries, and prices.

| Stock            | Orders      | Customer    | Country   | Price        | Costs                  | Categories |
|------------------|-------------|-------------|-----------|--------------|------------------------|------------|
| SKU-ID           | InvoiceNo   | CustomerID  | InvoiceNo | ID           | SKU                    | ID         |
| Description      | SKU         | InvoiceDate | Country   | Retail_Price | raw_material           | category   |
| 2020_units_sold  | InvoiceDate |             |           |              | factory_labor          |            |
| 2021_start_stock | Quantity    |             |           |              | factory_equipment_rent |            |
|                  | Country     |             |           |              | distribution           |            |
|                  |             |             |           |              | advertisement          |            |

# Let's practice!

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